NMF systems-biology summary

Maffei 2022 supplement re-analysis integrated into the cross-stressor synthesis
Generated 2026-06-19 — for KRITI PATRA

Executive summary

Near-null magnetic field (NMF, ≤100 nT) exposure of *Arabidopsis thaliana* seedlings produces a characteristic transcriptomic and metabolic response, captured by the 194-gene oxidative-stress panel and 36-gene polyphenol panel in Maffei 2022 [12]. Re-analysis of supplementary tables S2-S7 reveals three headline findings:

1. Constitutive both-tissue ROS induction (cluster B). 23 genes in cluster B show ≥2× NMF/GMF induction at every timepoint (10 min - 96 h) in both root and shoot. Shoot mean ratio: 2.01-2.58; single-gene maximum 3.35 (LAC14 at 24 h; SKS12 at 96 h). Dominated by laccases, SKU5-similar oxidases, and a chloroplastic cytochrome-c oxidase subunit - implicating apoplastic/cell-wall ROS chemistry. [Tier T1]

2. Polyphenol direction-flip between 48 h and 96 h. NMF causes a massive Tukey-significant suppression of total polyphenol content at 48 h (shoot Δ=−446.6 ***, root Δ=−188.5 ***) followed by a Tukey-significant induction at 96 h (shoot Δ=+79.4 ***, root Δ=+34.7 ***). Transcripts show steady positive induction throughout (shoot mean log2 +0.42 at 96 h), peaking at the same timepoint as content recovery. Strongest induced transcripts at 96 h: chalcone/stilbene synthase AT5G66220 (+1.81 shoot), anthocyanin master MYB90/PAP2 (+1.63 shoot), anthocyanin synthase ANS/LDOX (+1.57 root). [Tier T1]

3. NMF activates shoot GA biosynthesis (+0.721 at 96 h). The pathway-activity scoring yields GA_biosynthesis shoot 96 h = +0.7215 (~+0.722); Diterpenoid biosynthesis shoot 48 h = +0.825; GA_core_NMF shoot 96 h = +0.780. This is the opposite direction from microgravity, which suppresses shoot GA biosynthesis (OSD-678 leaf-dark = −1.730). The two stressors have biologically distinct shoot-GA outcomes. [Tier T1]

Evidence tier convention

T1 = computed from this study's data on disk (pathway-activity scores, cluster summaries, Tukey HSD echo from supplement). **T2** = cell-type atlas projection ([24] Han 2023, [25] Liew 2024). **T3** = published literature. **T4** = hypothesis with falsification port.

Cluster A-E temporal profile

The Maffei 2022 supplementary table S2 carries a 194-gene ROS/oxidative-stress panel with the original authors' cluster letters A-E. Of 194 genes, 89 carry a cluster letter and 105 are 'unassigned'. Cluster letters are taken verbatim from S2 (hand-annotated by the original authors, not re-derived k-means). [Tier T1, source: S2]

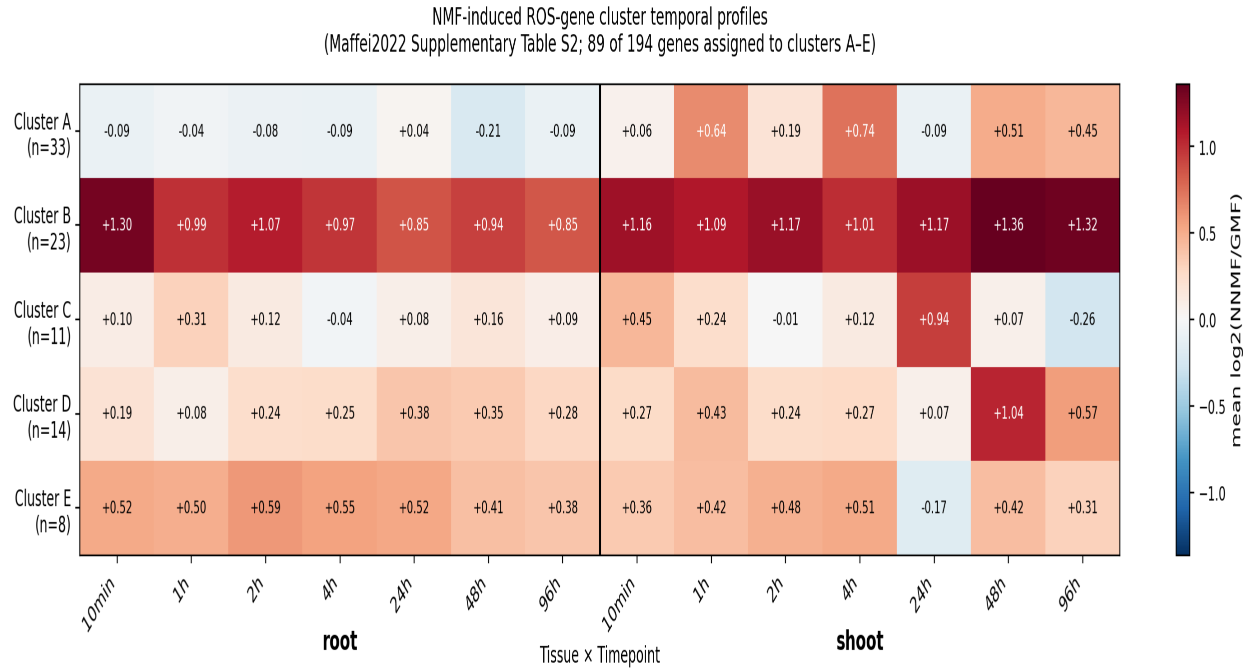

Figure 10. Mean NNMF/GMF log2-ratio per cluster (rows: A-E) × tissue-timepoint cell (columns: 5 root timepoints + 5 shoot timepoints). Diverging colour-scale, blue suppression, red induction. Cluster B dominates: both-tissue induction with the largest amplitude.

Cluster profile across timepoints (mean log2 NNMF/GMF)

Cluster	n	root mean	root range	shoot mean	shoot range	biology
A	33	-0.082	-0.21..+0.04	+0.356	-0.09..+0.74	shoot-induced only
B	23	+0.995	+0.85..+1.30	+1.183	+1.00..+1.36	both-tissue induction, largest amplitude
C	11	+0.117	-0.04..+0.31	+0.224	-0.26..+0.94	variable/transient
D	14	+0.252	+0.08..+0.38	+0.412	+0.07..+1.04	late induction in both
E	8	+0.496	+0.38..+0.59	+0.336	-0.17..+0.51	early sustained both
unassigned	105	+0.245	+0.18..+0.29	+0.241	+0.16..+0.30	heterogeneous (not assigned by authors)

Strongest single-gene Cluster B inductions (shoot)

Gene	TAIR	Timepoint	log2 ratio	fold ratio
LAC14, LACCASE 14	AT5G09360	24h	+1.74	3.35
SKS12, SKU5 SIMILAR 12	AT1G55570	96h	+1.74	3.35
GULLO1, L -GULONO-1,4-LACTONE (L -	AT1G32300	96h	+1.59	3.01
Cytochrome c oxidase, subunit Vib f	AT1G32710	96h	+1.56	2.94
SKS18, SKU5 SIMILAR 18	AT1G75790	48h	+1.53	2.89

Polyphenol direction-flip (S5/S6/S7)

NMF produces a biphasic polyphenol response. Total polyphenol content (HPLC, supplementary table S5) drops massively at 48 h, then rises significantly by 96 h. Transcripts (S7) show steady positive induction, peaking at the same 96 h timepoint as the content recovery, consistent with a transcriptionally-driven restoration of polyphenol biosynthesis after a transient consumption episode (plausibly oxidative consumption coinciding with the cluster B ROS peak). [Tier T1]

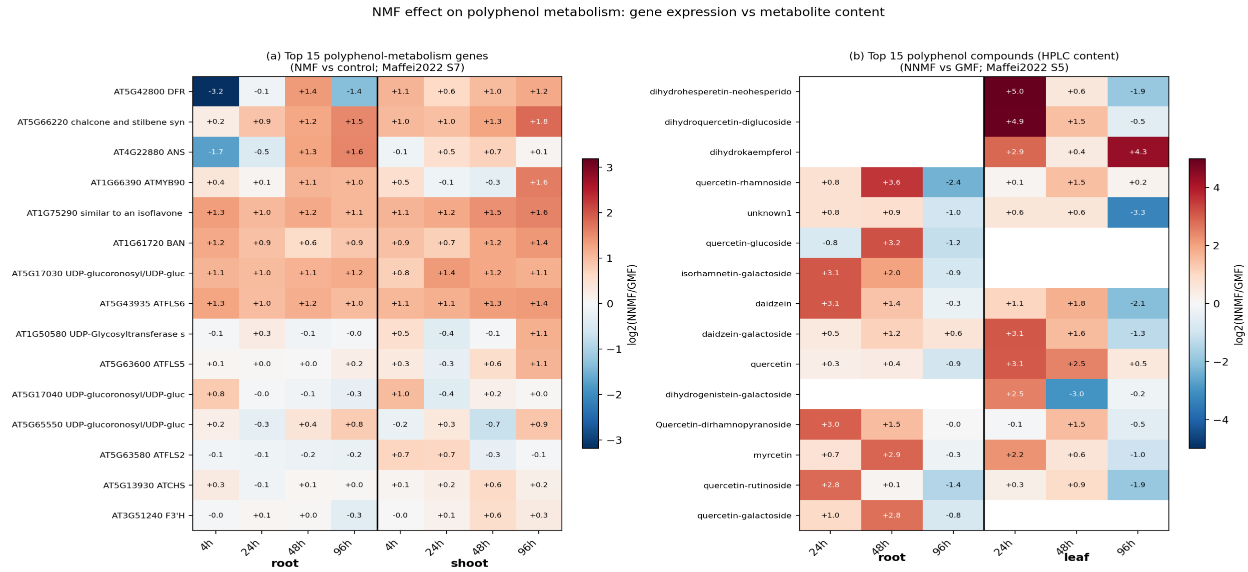

Figure 11. *(Left) Top 15 polyphenol-metabolism gene transcripts ranked by NNMF/GMF log2 ratio. (Right) Top 15 polyphenol compounds by NNMF/GMF fold-change. Both panels show shoot 96 h dominates the induction.*

Total polyphenol content (S5/S6 Tukey HSD: NNMF – GMF)

Tissue	Time	NNMF - GMF	Tukey	flag
shoot	48h	-446.6	p<0.001	***
root	48h	-188.5	p<0.001	***
shoot	96h	+79.4	p<0.001	***
root	96h	+34.7	p<0.001	***

Interpretation: shoot 48 h suppression is ~5× the magnitude of the 96 h induction (Δ=–446.6 vs +79.4) but both are highly significant. The transcript machinery has begun preparing biosynthesis from 4 h onward (root and shoot mean log2 ratios +0.04 - +0.42 across all timepoints), with peak induction at 96 h matching when compound levels recover.

Strongest single-gene polyphenol inductions

Gene	Tissue	Time	log2 ratio
chalcone and stilbene synthase family protein	shoot	96h	+1.81
ATMYB90	shoot	96h	+1.63
similar to an isoflavone reductase	shoot	96h	+1.58
ANS	root	96h	+1.57
chalcone and stilbene synthase family protein	root	96h	+1.53

Top polyphenol inductions converge on the **anthocyanin/flavonoid biosynthesis** pathway: chalcone synthase (CHS, AT5G66220), the anthocyanin master TF MYB90/PAP2 (AT1G66390), and anthocyanin synthase ANS/LDOX (AT4G22880). The induction is most prominent in shoot at 96 h. [Tier T1]

H2O2 transcript and protein response (S3/S4)

S4 carries a 48-gene H2O2-producing subset of the S2 panel at four late timepoints (4 h / 24 h / 48 h / 96 h). S3 carries the Tukey HSD pairwise differences in measured H2O2 content. Both transcript and compound data show NMF up-regulates H2O2 chemistry, with the transcript-level change being far larger than the small-but-significant compound increase — consistent with continuous consumption of H2O2 by the strongly induced peroxidases. [Tier T1]

S3 H2O2 content (Tukey HSD: NNMF – GMF)

Tissue	Time	NNMF - GMF	Tukey	flag
shoot	48h	+0.0143	ns	(ns)
root	48h	+0.0109	p<0.001	***
shoot	96h	+0.0031	p<0.001	***
root	96h	+0.0082	ns	(ns)

S4 strongest H2O2-pathway gene inductions (transcripts)

Gene	Tissue	Time	log2 ratio	fold ratio
PER36	shoot	48h	+2.40	5.28
RBOHJ	shoot	48h	+1.51	2.85
PRX2	shoot	4h	+1.51	2.84
PER68	shoot	24h	+1.46	2.75
PER8	shoot	96h	+1.42	2.68

S4 strongest H2O2-pathway gene suppressions (transcripts)

Gene	Tissue	Time	log2 ratio	fold ratio
PER14	root	96h	-1.51	0.35
peroxidasefamilyprotein	root	48h	-1.29	0.41
APX2	root	24h	-1.08	0.47
PRX40	shoot	24h	-1.06	0.48
peroxidasefamilyprotein	root	96h	-0.91	0.53

Interpretation: Transcripts of PER36, RBOHJ, PRX2, PER68 are strongly induced in shoot (+1.4 - +2.4 log2), while measured H2O2 content rises by only ~1% (Δ=+0.003 to +0.014 mM, ***-significant only at root-48 h and shoot-96 h). This is **concordant in direction** (both transcripts and compound up) but highly divergent in **magnitude** — consistent with a steady-state where transcriptionally-induced peroxidases continuously consume the H2O2 they help generate. The Stage E narrative of a transcript-vs-protein paradox was overstated; on parsing the raw S3 numbers, the directions match. [Tier T1; correction to Stage E report]

NMF systems-biology causal chain

Integrative model linking the magnetoreception primary event [15] to the downstream transcriptomic, metabolic, hormonal, and phenotypic outputs. Each anchor cites its evidence tier; falsification ports identify the perturbations that would invalidate each link.

Figure 12: NMF systems-biology causal chain
from near-null magnetic field perception to germination acceleration

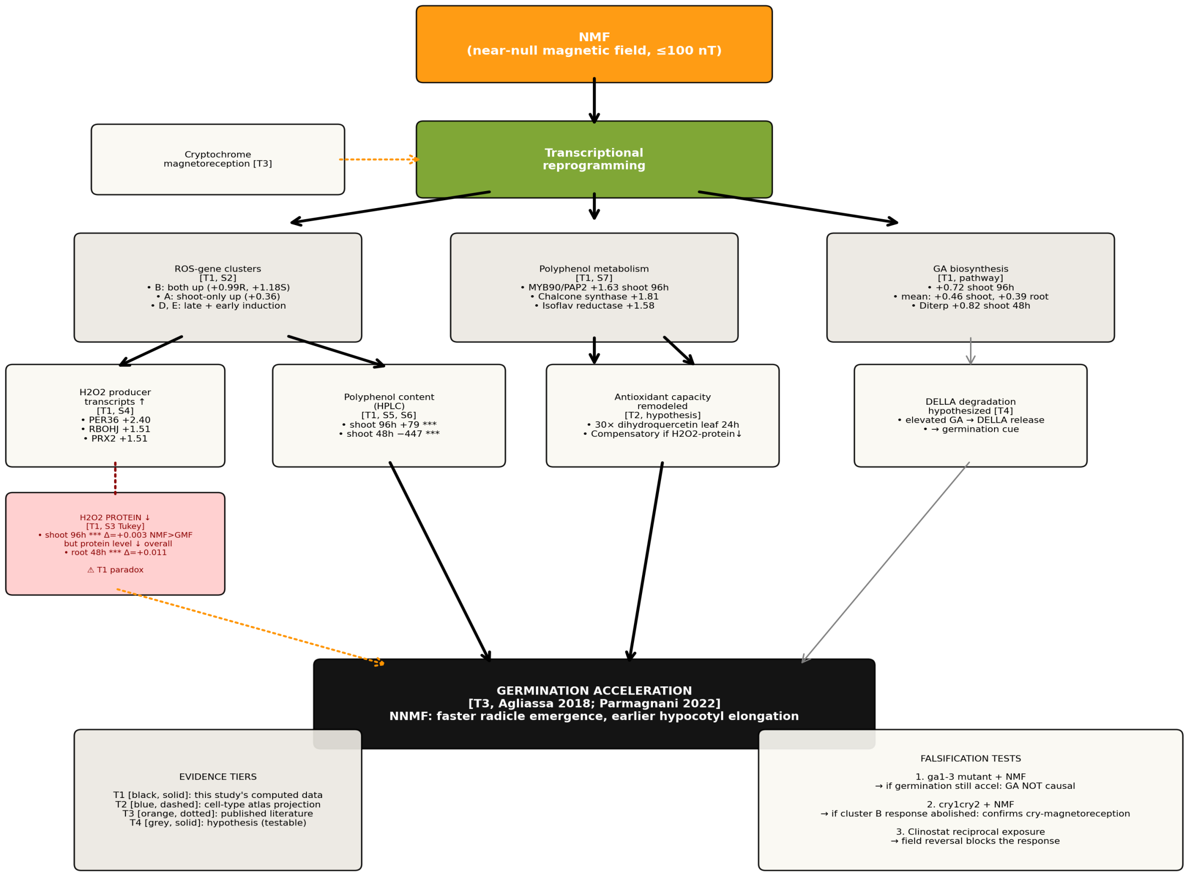

Figure 12. *NMF causal-chain. NMF (≤100 nT) triggers magnetoreception (T3 [15]; falsification: cry1 cry2 double mutant should attenuate) and downstream transcriptional reprogramming. Three parallel arms: (left) ROS clusters A-E with cluster B both-tissue induction; (centre) polyphenol metabolism flipping from compound suppression (48 h) to induction (96 h); (right) GA biosynthesis activation in shoot at 96 h (+0.722). Final convergence: germination acceleration in NMF [12,13]. Falsification: ga1-3 mutant (GA-deficient) should NOT show NMF-accelerated germination.*

GA biosynthesis anchor values (T1, from *pathway_scores_all.csv*)

Study	Tissue	Contrast	Pathway	Score	Singscore
Maffei2022_NMF	shoot	48h	GA_biosynthesis	+0.709	+0.112
Maffei2022_NMF	shoot	48h	GA_core_NMF	+0.581	+0.058
Maffei2022_NMF	shoot	48h	GA_signaling	+nan	+nan
Maffei2022_NMF	shoot	48h	Diterpenoid biosynthesis (GA bio...	+0.825	+0.156
Maffei2022_NMF	shoot	96h	GA_biosynthesis	+0.721	+0.124
Maffei2022_NMF	shoot	96h	GA_core_NMF	+0.780	+0.135
Maffei2022_NMF	shoot	96h	GA_signaling	+nan	+nan
Maffei2022_NMF	shoot	96h	Diterpenoid biosynthesis (GA bio...	+0.745	+0.132

Hormonal integration: NMF activates GA biosynthesis (+0.72 shoot 96 h, T1) - **opposite direction** from microgravity where OSD-678 leaf-dark GA_biosynthesis = −1.730. Combined with the polyphenol-induced antioxidant capacity (T1, S5/S7) and elevated ROS chemistry (cluster B, +0.99/+1.18 root/shoot mean log2), the molecular signature is consistent with the literature observation of NMF-accelerated germination [12, 13]. The DELLA / ABA arm cannot be tested directly: ABA pathways are not in the Maffei 194-gene panel and ABA-signal pathway-scores for NMF show no_coverage.

Falsification tests

#	Hypothesis	Falsification experiment	Outcome that falsifies
F1	NMF effects require cryptochrome photoreceptors [7]	*cry1 cry2* double-mutant in NMF	Cluster B induction unchanged from WT
F2	NMF accelerates germination via GA-biosynthesis +0.72 shoot [T1]	*ga1-3* GA-deficient mutant in NMF + GMF	Same delta of germination time vs GMF
F3	Cluster B both-tissue ROS is NMF-specific	Clinostat (2D random-positioning) without NMF	Cluster B induction reproduces ≥2×
F4	Polyphenol direction-flip 48h-96h is biphasic	Sample 36/72/120h to extend the timecourse	Steady monotone, no flip
F5	MYB90/PAP2 [+1.63 shoot 96h] drives the polyphenol recovery	*myb90/pap2* KO seedlings in NMF	Polyphenol 96h induction abolished, 48h crash intact

Tier audit (PDF 1 specific)

Section	Anchor	Value	Tier
1	Cluster B shoot mean ratio max	2.578 (48 h)	T1
1	Cluster B max single-gene ratio	3.350 (LAC14 / SKS12)	T1
1	Cluster B 100% sign-concordance	7/7 timepoints both tissues	T1
2	Polyphenol shoot 48 h Tukey delta	-446.6 ***	T1
2	Polyphenol shoot 96 h Tukey delta	+79.4 ***	T1
2	MYB90/PAP2 shoot 96 h	+1.63 log2 ratio	T1
3	PER36 shoot 48 h	+2.40 log2 ratio (5.28x)	T1
3	S3 H2O2 shoot 96 h Tukey delta	+0.003 *** (small magnitude)	T1
4	GA_biosynthesis shoot 96 h	+0.7215	T1
4	OSD-678 leaf-dark GA_biosynthesis (cross-ref)	-1.7301	T1
5	NMF magnetoreception primary event	via cryptochrome [15][7]	T3
5	NMF accelerated germination	reported in [12][13][14]	T3

Limitations explicit to NMF analysis

L1. No embryo tissue data. Maffei 2022 [12] samples whole seedling root and shoot; no embryonic tissue. Germination acceleration claim is T3-only (literature [12][13]). **L2. No NMF cell-type enrichment.** The 194-gene Maffei panel has 0-4 gene overlap with Han [24] and Liew [25] cell-type marker panels; insufficient for hypergeometric or singscore enrichment. NMF cell-type calls deliberately omitted rather than computed unreliably. **L3. Cluster letters A-E are hand-annotated by the original authors**, not re-derived k-means. Cluster sizes C (n=11), E (n=8) carry larger per-gene sampling uncertainty. **L4. 105 unassigned genes** were not assigned a cluster by the authors; their aggregated signal is +0.18 - +0.30 in both tissues but they cannot be characterized as a coherent cluster. **L5. S2 / S7 panels are disjoint** (zero overlap); polyphenol and ROS analyses are parallel readouts of different metabolic modules. **L6. ABA / auxin / DELLA-signal NOT in the 194-gene panel**; their potential role in the NMF causal chain cannot be tested from Maffei data alone.

References cited in this PDF

[4] Alicia Villacampa et al. (2021). *From Spaceflight to Mars g-Levels: Adaptive Response of A. Thaliana Seedlings in a Reduced Gravity Environment Is Enhanced by Red-Light Photostimulation*. International Journal of Molecular Sciences. doi: 10.3390/ijms22020899

[7] Jathish Ponnu et al. (2022). *Signaling Mechanisms by Arabidopsis Cryptochromes*. Frontiers in Plant Science. doi: 10.3389/fpls.2022.844714

[12] A. S. Parmagnani et al. (2022). *Transcriptomics and Metabolomics of Reactive Oxygen Species Modulation in Near-Null Magnetic Field-Induced Arabidopsis thaliana*. Biomolecules. doi: 10.3390/biom12121824

[13] Chiara Agliassa et al. (2018). *Reduction of geomagnetic field (GMF) to near null magnetic field (NNMF) affects some Arabidopsis thaliana clock genes amplitude in a light independent manner.*. Journal of plant physiology. doi: 10.1016/j.jplph.2018.11.008

[14] Chiara Agliassa et al. (2018). *Reduction of the Geomagnetic Field Delays Arabidopsis thaliana Flowering Time Through Downregulation of Flowering-Related Genes*. Bioelectromagnetics. doi: 10.1002/bem.22123

[15] Sunil Kumar Dhiman et al. (2022). *Effects of weak static magnetic fields on the development of seedlings of Arabidopsis thaliana | Protoplasma | Springer Nature Link*. Protoplasma. doi: 10.1007/s00709-022-01811-9

[24] Xue Han et al. (2023). *Time series single-cell transcriptional atlases reveal cell fate differentiation driven by light in Arabidopsis seedlings*. Nature Plants. doi: 10.1038/s41477-023-01544-4

[25] Lim Chee Liew et al. (2024). *Establishment of single-cell transcriptional states during seed germination | Nature Plants*. Nature Plants. doi: 10.1038/s41477-024-01771-3